Independent Sets in Regular Hypergraphs at Both Extremes

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Abstract

Balogh, Bollobás, and Narayanan conjectured that a particular quasi-bipartite construction maximizes the number of independent sets in a uniform regular hypergraph. Sarantis, Tetali, and Zheng proposed corresponding partition-function and coefficient inequalities. We prove these inequalities in intervals at both ends of their parameter ranges. For every fixed uniformity at least three and degree at least two, the partition-function bound holds at all sufficiently small positive activities. For degree at least three it also holds at all sufficiently large activities, with an exponential penalty for vertices outside the conjectured extremal components. We prove coefficient bounds throughout a positive linear interval near zero and, for degree at least three, another near the maximum possible independent-set size. The proofs combine local overlap deficits and analytic comparison at small activity with a degree-sensitive matching entropy bound and regular completion in the dual hypergraph at large activity. Both ends admit stability conclusions. In degree two, a weighted regional argument proves the partition-function bound at every nonnegative activity for all uniformities, with an exact equality characterization.

1 Introduction

An independent set in a hypergraph is a vertex set containing no edge. All hypergraphs in this paper are finite and simple, except for explicitly labeled dual multihypergraphs, and have positive order unless the empty case is mentioned. Let $i_k(H)$ count the independent k-sets of H, and put $Z_H(\lambda) = \sum_{k\ge 0} i_k(H)\lambda^k$. The parameter $\lambda \ge 0$ is the activity. Thus $Z_H(1)$ counts all independent sets, whereas large activity favors sets close to the largest possible size.

The reference hypergraph $H_{r,d}$ has d disjoint sets of size $r-1$, called cores, and d further vertices, called pages. Its edges are all unions of one core and one page. It has rd vertices and is d-regular. Its partition function is

$$P(\lambda) := Z_{H_{r,d}}(\lambda) = (1+\lambda)^{(r-1)d} + \big((1+\lambda)^d - 1\big)\big((1+\lambda)^{r-1} - \lambda^{r-1}\big)^d. \tag{1}$$

The first term counts sets containing no page; every other independent set omits at least one vertex from each core.

Balogh, Bollobás, and Narayanan [1, Conjecture 1.1] conjectured that $Z_H(1) \le P(1)^{n/(rd)}$ for every n-vertex r-uniform d-regular hypergraph. They proved this for quasi-bipartite hypergraphs. Sarantis, Tetali, and Zheng [8, Conjectures 1.2(ii) and 1.7] proposed the stronger inequalities

$$Z_H(\lambda) \le P(\lambda)^{n/(rd)}, \qquad i_k(H) \le i_k(H^n_{r,d}), \tag{2}$$

where the first is asserted for every $\lambda \ge 0$, and the second assumes $rd \mid n$ and uses the disjoint union $H^n_{r,d}$ of $n/(rd)$ reference components. Their asymptotic results impose a condition on the

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twin quotient or exclude cross-edges. They also settle degree two for odd uniformity and for uniformity four, and note that these cases extend to all activities. Ferudun [4] subsequently established the degree-two counting inequality at $\lambda = 1$ for all uniformities in a preprint. We extend the regional argument to all activities, completing the remaining even uniformities at least six, and characterize equality. For larger degrees, our interval results impose none of the preceding structural assumptions.

Let $b(H)$ count the vertices not belonging to a connected component isomorphic to $H_{r,d}$. When $rd \mid n$, let $\operatorname{dist}(H, H^n_{r,d})$ be the minimum size of the symmetric difference of edge sets over all copies of $H^n_{r,d}$ on $V(H)$. We use the vertex count at large activity and the edge-edit distance at low density.

Theorem 1.1. *Fix $r \geq 3, d \geq 2$. There exists $\lambda_-(r,d) > 0$ such that every n-vertex r-uniform d-regular hypergraph satisfies the first inequality in (2) for $0 \leq \lambda \leq \lambda_-$. At positive activity in this interval, equality holds precisely for disjoint unions of $H_{r,d}$.*

If $d \geq 3$, there are $\lambda_+(r,d) > 1$ and $\eta(r,d) > 0$ such that, for every $\lambda \geq \lambda_+$,

$$Z_H(\lambda) \leq P(\lambda)^{n/(rd)} \exp(-\eta b(H)/8). \tag{3}$$

Neither assertion requires $rd \mid n$ or a lower bound on n.

Degree two admits a complete activity comparison. The formula below is (1) with $d = 2$.

Theorem 1.2. *For every $r \geq 2$, every finite simple r-uniform 2-regular hypergraph H on n vertices, and every $\lambda \geq 0$,*

$$Z_H(\lambda) \leq \left[(1+\lambda)^{2r-2} + \big((1+\lambda)^2 - 1\big)\big((1+\lambda)^{r-1} - \lambda^{r-1}\big)^2\right]^{n/(2r)}.$$

For $\lambda > 0$, equality holds if and only if H is a disjoint union of copies of $H_{r,2}$. No divisibility assumption on n is required.

An exact transversal meets every edge in exactly one vertex. Let $T(H)$ count these sets, and set $A = 1 + (r-1)^d$. The high-activity argument begins with a stable bound on this count.

Theorem 1.3. *For $r \geq 3, d \geq 3$, the constant η in Theorem 1.1 can be chosen so that*

$$T(H) \leq A^{n/(rd)} \exp(-\eta b(H)) \tag{4}$$

for every n-vertex r-uniform d-regular hypergraph. If $r \mid n$, then $T(H) = i_{(r-1)n/r}(H)$. The unpenalized bound has equality precisely for disjoint unions of $H_{r,d}$.

The complement of an independent set is a transversal. Counting incidences gives $d|V(H) \setminus I| \geq nd/r$, hence $|I| \leq (r-1)n/r$. Equality holds exactly when the complement is an exact transversal. We also obtain coefficient inequalities in intervals of positive linear length at both ends.

Theorem 1.4. *Fix $r \geq 3, d \geq 2$. There exist $\alpha_-(r,d) > 0$ and $N(r,d)$ such that, whenever $n \geq N$ and $rd \mid n$, every n-vertex r-uniform d-regular hypergraph satisfies*

$$i_k(H) \leq i_k(H^n_{r,d}) \qquad (0 \leq k \leq \alpha_- n).$$

For $2r \leq k \leq \alpha_- n$, equality holds precisely when $H \cong H^n_{r,d}$. For every fixed $\tau \in (0, \alpha_-]$, there is $c = c(r,d,\tau) > 0$ such that, for n sufficiently large and $\tau n \leq k \leq \alpha_- n$,

$$i_k(H) \leq i_k(H^n_{r,d}) \exp(-c \operatorname{dist}(H, H^n_{r,d})). \tag{5}$$

If $d \geq 3$, there is also $\alpha_+(r,d) > 0$ such that, for every $n > 0$ divisible by rd and every integer $(r-1)n/r - \alpha_+ n \leq k \leq (r-1)n/r$,

$$i_k(H) \leq i_k(H^n_{r,d}) \exp(-\eta b(H)/16). \tag{6}$$

Equality in this latter interval is possible only when $H \cong H^n_{r,d}$.

All constants depend only on the fixed pair (r, d) and are not optimized. The low-density result requires sufficiently large n, whereas the high-density result holds for every admissible order. For small orders, the latter interval may contain only the maximum possible size. For degrees at least three, these theorems leave intermediate activities and the general counting problem at $\lambda = 1$ unresolved. The all-activity result in degree two does not imply comparison of every coefficient.

For $(r, d) = (3, 3)$, a companion manuscript [6] proves the counting bound and the partition-function comparison on $[2/3, 5/3]$, a middle interval not supplied by the endpoint results here.

At small activity, local overlap counts control the edges outside complete books. A regular repair and a complex zero-free estimate then control all later logarithmic coefficients by the same nonnegative deficits. At large activity, maximum twin classes give a two-state decomposition. A degree-sensitive version of Luria's random-order matching entropy argument [7] supplies a strict loss outside reference components. Regular completion in the dual propagates it from exact transversals to near-minimum covers.

The passage from stability to coefficient bounds has precedents in the graph results of Davies, Jenssen, Perkins, and Roberts [3] and Davies, Jenssen, and Perkins [2]. Here the low-density input is a layered analytic deficit, and the Fourier estimate concerns only powers of the fixed reference polynomial. At the other end, a weighted double count handles convolution with many reference components. Sections 2 and 3 prove the activity bounds, including the complete degree-two case, and Section 4 proves the coefficient intervals. All logarithms are natural.

2 Local deficits and small activity

A complete d-book consists of $C \cup \{p_1\}, \dots, C \cup \{p_d\}$, where $|C| = r - 1$ and the pages are distinct. In a hypergraph of maximum degree d, its core vertices have no other incident edges. For $d \geq 2$, distinct complete books have disjoint cores: an intersection vertex would force their incident edge sets, and hence the intersections of those edge sets, to agree. A page cannot lie in another core, since its book edge would then contain that entire disjoint core. A regular hypergraph consisting entirely of book edges is a *full-book hypergraph*. It comes from a simple d-regular bipartite graph by replacing each vertex in one side with $r - 1$ twins, that is, vertices with identical incident edge sets.

Write $m = nd/r$ and $q = d - 1$. For an $(r + 1)$-set X, put $j_X = e(H[X])$ and define

$$C_{r+1}(H) = \sum_{|X|=r+1} (j_X - 1)_+, \qquad \delta = mq/2 - C_{r+1}(H).$$

Counting edge multiplicities in $(r + 1)$-sets gives

$$i_{r+1}(H) = \binom{n}{r+1} - m(n - r) + C_{r+1}(H). \tag{7}$$

Lemma 2.1. *For $r \geq 3, d \geq 2$, one has $\delta \geq 0$. If $r \geq 4$, at most 4δ edges lie outside complete d-books.*

Proof. Let L be the graph on $E(H)$ joining pairs whose intersection has size $r-1$. Put $\rho_e = d_L(e)$, $B = e(L)$,

$$T = \sum_X \binom{j_X}{3}, \qquad R = \sum_{X : j_X \geq 2} \binom{j_X - 1}{2}.$$

Then $B = \sum_X \binom{j_X}{2}$ and $C_{r+1} = B - R$. Each vertex of X lies in at least $j_X - 1$ edges of $H[X]$, so $j_X \leq d + 1$. It follows that $R \geq 3T/(d + 1)$.

For adjacent $e = C \cup \{x\}$ and $f = C \cup \{y\}$, let c count edges containing C, and put $X = e \cup f$. Let U, V count neighbors of e, f, respectively, that neither contain C nor lie in X, and put

$W = j_X - 2$. Every additional common neighbor contains C or lies in X. The degree budget on C gives $(r-2)(U+V+W) \le (r-1)(d-c)$. Since $\rho_e + \rho_f = 2c - 2 + U + V + 2W$, it follows that

$$s_{ef} := 2q + j_X - 2 - \rho_e - \rho_f = 2(d-c) - (U+V+W) \ge 0. \tag{8}$$

Summation and Cauchy's inequality give $4B^2/m \le \sum_e \rho_e^2 \le 2qB + 3T$. If $d \ge 3$ and $B = mq/2 + x$ with $x > 0$, then $3T \ge 2qx + 4x^2/m$ and

$$B - R \le mq/2 + \frac{3-d}{d+1}x - \frac{4x^2}{(d+1)m} < mq/2.$$

If $B \le mq/2$, the result follows from $R \ge 0$. For $d = 2$, two neighbors of one edge would force a vertex of degree three, so L has maximum degree one. This proves $\delta \ge 0$.

For $d \ge 3$, set $V_0 = \sum_e (\rho_e - q)^2$ and $E_0 = \sum_{ef \in E(L)} s_{ef}$. Direct expansion gives

$$V_0 + E_0 + \sum_X (2q - j_X)\binom{j_X - 1}{2} = 2q\delta. \tag{9}$$

The last summands are nonnegative. For $r \ge 4$, the degree budget strengthens (8) to $s_{ef} \ge (r-3)(d-c)/(r-2)$. Each complete book forms an isolated K_d in L. Every other adjacent pair has $c \le d-1$, so its integral slack is at least one. If b edges lie outside books, then $E_0 \ge \frac{1}{2}\sum_{e \text{ outside books}} \rho_e$. Since $(\rho - q)^2 + \rho/2 \ge q/2$ for integral $\rho \ge 0$, we obtain $qb/2 \le V_0 + E_0 \le 2q\delta$. For $d = 2$, the exceptional edges are the isolated vertices of L, numbering $m - 2B = 2\delta$. □

For triples the first deficit alone is insufficient. The following five-vertex calculation supplies the additional structural information.

Lemma 2.2. *For $r = 3, d \ge 2$, there exists $\xi \ge 0$, with $\xi = O_d(n)$, such that at most $1000d^2(\delta + \xi)$ edges lie outside complete books. For any full-book hypergraph H_0 with the same n, d,*

$$\log Z_{H_0}(z) - \log Z_H(z) = \delta z^4 + (\xi - 200d^2\delta)z^5 + O(z^6) \tag{10}$$

as a formal power series.

Proof. Use the notation of Lemma 2.1. Now $R = T - \#K_4^{(3)}$, so $T \le 4R/3$. Equation (9) yields $V_0 \le 2q\delta$ and, for $d \ge 4$,

$$R \le q\delta/(d-3). \tag{11}$$

For each five-set S, let G_S be the graph with edges $S \setminus e$, $e \in H[S]$. For a nonempty graph G on at most five vertices, let $v_+(G)$ count nonisolated vertices and $M_2(G)$ count unordered pairs of disjoint edges. Put

$$D(G) = e(G) - v_+(G) + 1 + M_2(G), \qquad P_3(G) = \sum_{uv \in E(G)} (d_G(u) - 1)(d_G(v) - 1),$$

and set both quantities to zero for an empty graph. Write $J(G) = D(G) - P_3(G)/3$ and $D_5 = \sum_S D(G_S)$. If $p(G)$ counts unoriented three-edge paths and $t(G)$ counts triangles, then $P_3 = p + 3t$.

If G is K_4-free, then $J(G) \ge 0$. Let c count its nonempty components and M_{cross} its pairs of edges from different components. Each pair of disjoint edges in one component can be joined into at most three three-edge paths, since four joining edges would form a K_4. Thus $p \le 3(M_2 - M_{\text{cross}})$. Also $t \le e - v_+ + c$. This follows inductively by deleting vertices of degree at most two. The only remaining K_4-free graph with minimum degree at least three on at most five vertices is a four-cycle with a universal fifth vertex, where both sides equal four. Since $M_{\text{cross}} \ge c - 1$, these inequalities prove $P_3 \le 3D$.

Call S clean if $\Delta(G_S) \le 2$, and put $J_{\mathrm{cl}} = \sum_{S \text{ clean}} J(G_S)$. The clean graphs containing a three-edge path have nontrivial component P_4, P_5, C_4, or C_5. Their respective pairs (D, p) are $(1,1), (3,2), (3,4), (6,5)$. Thus their total path count is at most $3J_{\mathrm{cl}}$.

Let W count induced three-vertex paths of L, with unordered ends. An edge outside books either has $\rho_e \neq q$, is the middle of such a path, or belongs to a triple of edges whose union has four vertices. Otherwise its closed neighborhood is a K_d, and its members share a pair: given abc, abx, a common neighbor either contains ab or lies in $\{a, b, c, x\}$. Therefore $b \le V_0 + W + 3T$. Every induced path of L has a five-vertex union and corresponds to a three-edge path in G_S. A nonclean S contains a four-set Q with at least three hyperedges. If its complement graph has a three-edge path, some hyperedge contains the outside vertex and two points of Q. There are at most $6d$ such outside vertices for each Q, at most R choices of Q, and at most 60 paths in a five-vertex graph. Hence

$$b \le 2q\delta + 3J_{\mathrm{cl}} + (360d + 4)R. \tag{12}$$

Set $D_0 = mq(d-2)/6$, $\zeta = 2R + D_5 - D_0 - 6\delta$, and $\xi = \zeta + 200d^2\delta$. To sum P_3, choose an edge and two of its neighbors in L. Their union has four or five vertices. Four-vertex unions contribute $3T$, whereas the choices with five-vertex union are exactly those counted by $P_3(G_S)$. Thus $\sum_S P_3(G_S) = \sum_e \binom{\rho_e}{2} - 3T$. Together with $\sum_e (\rho_e - q) = 2(R - \delta)$, this gives

$$\zeta = -\frac{2d+15}{3}\delta + \frac{2d+3}{3}R + \frac{V_0}{6} - T + \sum_S J(G_S). \tag{13}$$

For $d \le 5$, a K_4 in G_S would produce a hypergraph vertex of degree six. Thus $J(G_S) \ge 0$, and (13) implies $\xi \ge \delta + R + J_{\mathrm{cl}}$, using $R = 0$ when $d = 2$.

For $d \ge 6$, a graph with $J < 0$ contains a K_4, with the fifth vertex joined to j of its vertices. Here $M_2 = 3 + 3j$, and direct substitution gives

j	0	1	2	3	4
D	6	9	13	17	21
P_3	24	30	43	63	90
J	-2	-1	$-4/3$	-4	-9

.

Each such five-set is counted in the preceding Q argument, so $\sum_S J(G_S) \ge J_{\mathrm{cl}} - 54dR$. Using (11) in (13) gives $\xi \ge \delta + J_{\mathrm{cl}}$. Since $R \le 2\delta$, substitution into (12) proves the stated bound on b.

To verify the coefficients, put $C_5 = \sum_{|S|=5} (e(H[S]) - 1)_+$. The definition of D gives the local identity

$$(e(H[S]) - 1)_+ = 2e(G_S) - v_+(G_S) + M_2(G_S) - D(G_S),$$

including when G_S is empty. A nonisolated vertex of G_S corresponds to a four-set containing an edge of H, obtained by deleting that vertex. A pair counted by $M_2(G_S)$ corresponds to two hyperedges intersecting in one vertex. Hence

$$\sum_S v_+(G_S) = (n-4)(m(n-3) - C_4(H)), \qquad \sum_S M_2(G_S) = n\binom{d}{2} - 2B.$$

Summing the local identity, using $\sum_S e(G_S) = m\binom{n-3}{2}$ and $B = mq/2 + R - \delta$, yields

$$C_5 = (n-3)mq/2 - (n-6)\delta - 2R - D_5.$$

For a full-book hypergraph, $\delta = R = 0$ and $D_5 = D_0$. Its fourth and fifth independent-set coefficient differences from H are therefore δ and $(n-6)\delta + 2R + D_5 - D_0$. Passing to logarithms subtracts $n\delta$ from the fifth difference, proving (10). The nonzero local corrections are supported on connected collections of boundedly many edges. Fixed degree gives $O_d(n)$ such collections, and hence $\xi = O_d(n)$. □

Lemma 2.3. *If $rd \mid n$ and b edges of H lie outside complete books, there is a full-book hypergraph H_0 on $V(H)$, with the same r, d, satisfying $|E(H) \triangle E(H_0)| \le 4rb$.*

Proof. The number b is divisible by d. Its edges avoid all existing cores, and their support has at least rb/d vertices. Choose $(r-1)b/d$ of them as b/d new cores. Delete the exceptional edges and all old book edges whose pages were chosen, at most $T \le rb$ edges in total.

After compressing cores, the remaining bipartite graph has sides of size $n/r \ge d$, maximum degree d, and deficiency T on each side. Add edges between nonadjacent deficient vertices whenever possible. Otherwise choose deficient vertices a, z on opposite sides; they are adjacent. The vertex z has a nonneighbor b on the other side, and b must be full-degree. Some neighbor w of b is not a neighbor of a. Replacing bw by aw, bz reduces deficiency by one. At most T such switches suffice. Re-expanding cores gives a regular simple H_0, and the symmetric difference is at most $4T$. □

Full-book graphs satisfy the desired inequality at every activity by the product-space Hölder inequality of Finner [5]. Its equality case identifies the extremizers below.

Lemma 2.4. *For a full-book hypergraph, $Z_H(\lambda) \le P(\lambda)^{n/(rd)}$ for every $\lambda \ge 0$. At positive activity, equality holds precisely for reference unions.*

Proof. Put $p = \lambda/(1+\lambda)$ and select vertices independently with probability p. Integrating over a core gives a factor 1 if none of its pages is selected and $c = 1 - p^{r-1}$ otherwise. Each page occurs in exactly d core factors. Finner's inequality gives

$$\frac{Z_H(\lambda)}{(1+\lambda)^n} \le \left((1-p)^d + [1-(1-p)^d]c^d\right)^{n/(rd)}.$$

Equation (1) identifies the right side. For $0 < p < 1$, if two cores sharing a page v have different page neighborhoods, choose w in only the first neighborhood and condition on exactly w being selected among the other pages. Their two factors as functions of v are (c, c) and $(1, c)$, not proportional, so Hölder at v is strict. Otherwise every compressed component is $K_{d,d}$, which gives the equality classification. □

Lemma 2.5. *Let G have maximum degree at most $d \ge 2$, with nonempty edges of arbitrary sizes, and activities $|z_v| \le 1/(4d)$. Then $Z_G(\boldsymbol{z}) \ne 0$. The ratio corresponding to requiring all vertices of U to be selected has modulus at most $\prod_{v \in U} 4|z_v|$. Changing one r-edge between two hypergraphs of maximum degree d changes their analytic logarithms by at most $2(4|z|)^r$ when all activities equal z.*

Proof. Induct on the vertex count, simultaneously for nonvanishing and all ratios. A singleton edge forces its vertex ratio to be zero. Otherwise write $Z_G = A + z_v B$, excluding or forcing v. Starting from the graph for A, obtain that for B by adding at most d nonempty shortened edges. The intermediate graphs have fewer vertices and maximum degree at most d. By induction each insertion multiplies by $1-u$, with $|u| \le 1/d$. Hence $|B/A| \le (1+1/d)^d$. Since $(1+1/d)^{d+1} < 4$, the denominator is nonzero and $|z_v B/(A + z_v B)| \le 4|z_v|$. Successive conditioning proves the joint bound; an empty shortened edge gives a zero numerator. An added r-edge multiplies by $1-u$ with $|u| \le (4|z|)^r \le 1/8$. Now $|\log(1-u)| \le 2|u|$ proves the last assertion for the branch vanishing at zero. □

Proof of the small-activity part of Theorem 1.1. First assume $rd \mid n$ and take the repair H_0. By Lemmas 2.1, 2.3, and 2.5, the function $f_0 = \log Z_{H_0} - \log Z_H$ is analytic on a fixed disk and has modulus $O_{r,d}(\delta)$ there for $r \ge 4$. Equation (7) and Cauchy's estimate give $f_0(z) = \delta z^{r+1} + O_{r,d}(\delta z^{r+2})$, uniformly in n, H. For $r = 3$, the same argument with Lemma 2.2 gives

$$f_0(z) = \delta z^4 + (\xi - 200d^2\delta)z^5 + O_d((\delta + \xi)z^6).$$

At sufficiently small positive λ, these are at least $\delta\lambda^{r+1}/2$ and $(\delta\lambda^4 + \xi\lambda^5)/2$, respectively. Lemma 2.4 finishes the comparison. Equality forces the structural deficits to vanish, making H full-book, and then forces reference components. For arbitrary n, take rd disjoint copies of H, apply the comparison, and take the rdth root. The component equality conclusion descends as well. □

For coefficient extraction we must also penalize full-book graphs that are not reference unions. The next lemma summarizes the required analytic control.

Lemma 2.6. *Fix $r \geq 3, d \geq 2$ and assume $rd \mid n$. There are nonnegative D_j, indexed by $\mathcal{J} = \{4,5,6\}$ if $r = 3$ and $\mathcal{J} = \{r+1, 2r\}$ otherwise, and constants $C, R > 0$ depending only on r, d, with the following properties. Put $\ell = 2r$, $D = \sum D_j$, $Q(z) = \sum D_j z^j$, and $f(z) = \log P(z)^{n/(rd)} - \log Z_H(z)$. Then $D \leq Cn$, $D = 0$ precisely for reference unions, and $\operatorname{dist}(H, H^n_{r,d}) \leq CD$. For $0 < \lambda \leq R$,*

$$\sum_{j \geq 0} j\,|[z^j](f - Q)|\lambda^j \leq C\lambda Q(\lambda). \tag{14}$$

More explicitly, for $r \geq 4$,

$$f = \delta z^{r+1} + \gamma z^{2r} + \sum_{j=r+2}^{2r} a_j z^j + u(z), \qquad |a_j| \leq C\delta,$$

and for $r = 3$,

$$f = \delta z^4 + \xi z^5 + \gamma z^6 - 200d^2\delta z^5 + \chi z^6 + u(z), \qquad |\chi| \leq C(\delta + \xi).$$

In both cases u is analytic on $|z| \leq 2R$ and $|u(z)| \leq CD|z|^{\ell+1}$.

Proof. Let B be the compressed bipartite graph of H_0, with sides of size $t = n/r$. Set $\gamma = td(d-1)^2/4 - c_4(B) \geq 0$. Each edge is in at most $(d-1)^2$ four-cycles, with equality forcing its component to be $K_{d,d}$. Thus at most 4γ edges are outside those components. Their side sizes are equal and divisible by d. Replacing them with $K_{d,d}$ components changes at most 8γ hyperedges and produces $H^n_{r,d}$ on the same vertices.

The polynomials Z_{H_0} and $Z_{H^n_{r,d}}$ agree below degree $2r$, and their difference at degree $2r$ is γ. Indeed, with $p = z/(1+z)$, inclusion–exclusion gives

$$\frac{Z_{H_0}(z)}{(1+z)^n} = \sum_{F \subseteq E(B)} (-1)^{|F|} p^{(r-1)a(F)+b(F)},$$

where $a(F), b(F)$ count incident core and page vertices. At degree at most $2r$, a nonempty F uses at most two cores, since three cores and one page already give degree $3(r-1)+1 > 2r$. The possible types are stars, two-edge matchings, three-edge paths, and four-cycles. All their counts except four-cycles depend only on t, d. Four-cycles first appear at degree $2r$ with positive sign. The first logarithmic difference is the same.

The first repair changes $O_{r,d}(\delta)$ edges for $r \geq 4$, or $O_d(\delta + \xi)$ for $r = 3$. Lemma 2.5, (7), and (10), followed by Cauchy's estimate, give the displayed expansions. Shrinking the zero-free radius by a fixed factor gives the bound on u. Set $(D_{r+1}, D_{2r}) = (\delta, \gamma)$ or $(D_4, D_5, D_6) = (\delta, \xi, \gamma)$ as appropriate. The two repairs yield the distance bound, and the local counts give $D = O_{r,d}(n)$.

Each polynomial error contributes at most $C\lambda Q(\lambda)$ to the weighted absolute coefficient sum. For the tail expand $u(z)/z^{\ell+1}$ on the disk of radius $2R$ and sum the geometric coefficient bounds, with the factor j. Its contribution is at most $CD\lambda^{\ell+1} \leq C\lambda Q(\lambda)$, taking $R \leq 1$. This proves (14). □

3 Large activity and degree two

We first fix $r \geq 3, d \geq 3$ and write $a = r - 1$. The degree-two case is treated at the end of this section. We separate twin classes of maximum possible size. The residual choices become perfect matchings in a dual multihypergraph. Retaining degree deficits in the matching bound makes the inequality strict outside reference components.

Lemma 3.1. *Let J be a labeled s-uniform multihypergraph, $s \geq 2$, with parallel-edge multiplicities at most q_0. If it has a perfect matching and vertex degrees D_v, then*

$$\log \operatorname{pm}(J) \leq \frac{1}{s} \sum_v \int_0^1 \log(q_0 + (D_v - q_0)u)\, du. \tag{15}$$

For $q_0 = a - 1$ and $D_v \leq r$, put

$$I = \int_0^1 \log(a - 1 + 2u)\, du, \quad \beta = \mathrm{e}^I, \quad \gamma_0 = \int_0^1 \frac{u}{a - 1 + 2u}\, du \geq \frac{1}{2r}.$$

Then $\log \operatorname{pm}(J) \leq |V(J)|I/s - (\gamma_0/s) \sum_v (r - D_v)$.

Proof. Choose a uniform perfect matching M and, independently, uniform priorities in $[0, 1]$ on $V(J)$. Visit vertices in decreasing priority order, revealing the matching edge at each uncovered vertex. Already covered vertices contribute zero conditional entropy. At an uncovered vertex v, the possible labels are among its incident edges avoiding all previously covered vertices. The entropy chain rule bounds $\log \operatorname{pm}(J)$ by the expected sum of the logarithms of these candidate counts.

To estimate this expectation, fix M, the priority t of v, and the event that v is first in its matching edge. This event has probability t^{s-1} and constrains only the other priorities in that edge. If its multiplicity is $h \leq q_0$, its h labels remain candidates. Every other incident edge meets another matching block. To remain a candidate, all s priorities in that block must be below t; these priorities are independent of the conditioning, so this has probability t^s. Linearity of expectation, without independence between candidate events, bounds the mean candidate count by

$$h + (D_v - h)t^s \leq q_0 + (D_v - q_0)t^s.$$

The inequality holds even when $D_v < q_0$, since the expression increases with h. Jensen's inequality, followed by averaging over M and t, therefore gives

$$\log \operatorname{pm}(J) \leq \sum_v \int_0^1 t^{s-1} \log(q_0 + (D_v - q_0)t^s)\, dt.$$

Substitute $u = t^s$ to obtain (15). Its integral is concave in the positive variable D_v; the tangent at r gives the degree-deficit bound. Finally $a - 1 + 2u \leq r$ gives $\gamma_0 \geq \int_0^1 u/r\, du = 1/(2r)$. A graph with no perfect matching satisfies the corresponding bound for the count trivially, and the empty vertex set has one empty matching. □

This is a finite-parameter form of Luria's random-order entropy argument [7]. Its proof needs neither linearity nor an asymptotic codegree condition.

Proof of Theorem 1.3. Vertices are twins when they have the same incident edges. Each twin class has size at most a, since an r-vertex class would force repeated edges. Call a class of size a a large core. Its d page neighbors are distinct singleton twin classes. In the bipartite incidence graph between large cores and pages, a component with h_i cores and p_i pages has $h_i \geq 1$, $p_i \geq d$, and $p_i \geq h_i$. If z original vertices remain outside these components, then $n = z + \sum_i (ah_i + p_i)$.

Each component has two possible states in an exact transversal. The core state selects one vertex of each core and no page, giving a^{h_i} choices. The page state selects all pages and no core, giving one choice. Put $q_i = p_i - h_i \geq 0$, and let Q_C, Q_P be its sums over the two states.

Delete the book edges and those already covered by selected pages. In a legal state the number of uncovered edges is $N' = nd/r - d\sum_i h_i - dQ_P$. Regard them as dual vertices. Retain a remaining original vertex as a labeled dual edge only when all its d incident original edges are uncovered. The resulting dual is d-uniform, has maximum degree r, and multiplicity at most $a-1$. The last assertion follows because the remaining original vertices contain no large core and we retain whole dual edges, never shortened ones. Residual exact transversals are in bijection with its perfect matchings.

Unselected core-state pages have dQ_C incidences with nonbook edges. Page-state choices cover dQ_P such edges, removing at most adQ_P of those incidences. The residual dual degree deficit is therefore at least $d\max(0, Q_C - aQ_P)$. Lemma 3.1 bounds its matching count by $\beta^{N'/d}\exp[-\gamma_0 \max(0, Q_C - aQ_P)]$.

Set $\zeta = \log a - I$. Expansion around a gives

$$\zeta = \sum_{j\geq 1} \frac{1}{2j(2j+1)a^{2j}}, \qquad 0 < \zeta \leq \frac{1}{6(a^2-1)} < a^{-2}.$$

Choose $\theta = (a^{-2} - \zeta)/r \in (0, \gamma_0)$, and put $\beta_* = \beta \mathrm{e}^{-r\theta} = a\mathrm{e}^{-1/a^2}$, so $1 < \beta_* < \beta < a$. Replace the preceding penalty by the weaker factor $\exp[-\theta(Q_C - aQ_P)]$. Summing over states, including illegal states only as additional positive terms, yields

$$T(H) \leq \beta^{z/r} \prod_i \left(a^{h_i} \beta_*^{q_i/r} + \beta_*^{-aq_i/r} \right). \tag{16}$$

Write $C = A^{1/d} > a$ and $L = \log A/(rd)$. The ratio of one factor to its reference contribution is

$$F(h,q) = (a/C)^h (\beta_*/C)^{q/r} + C^{-h}(\beta_*^{-a}/C)^{q/r}.$$

It strictly decreases with q, whose feasible minimum is $\max(0, d-h)$. If $h \geq d$, the inequality $a^h + 1 \leq (a^d+1)^{h/d}$ gives $F(h,0) \leq 1$, with equality only for $h = d$. If $1 \leq h \leq d$, then $F(h, d-h)$ is a sum of two exponentials in h, hence convex. One endpoint equals one, and the other satisfies

$$F(1, d-1) = \mathrm{e}^{-(d-1)/(ra^2)} \frac{1 + a^{-d}\mathrm{e}^{(d-1)/a^2}}{(1+a^{-d})^{(a+d)/(rd)}} < 1. \tag{17}$$

We check the strict inequality explicitly. For $a \geq 3$, discard the denominator and use $\log(1+x) \leq x$. It suffices that $r/(a\beta_*^{d-1}) \leq (d-1)/a^2$. The left side decreases with d, and the right side increases. At $d = 3$ it follows from $(a+1)\mathrm{e}^{2/a^2} \leq 2a$, which is implied by $\mathrm{e}^t \leq 1/(1-t)$ and $a^2 - a - 4 > 0$. For $a = 2$, one has $3/2 < \beta_* < 8/5$, and the same sufficient condition holds for $d \geq 4$. For $d = 3$, it remains to prove $2\beta_*^{2/3} + \beta_*^{-4/3} < 9^{5/9}$. The left side is below $2(11/8) + 7/12 = 10/3$, by $(8/5)^2 < (11/8)^3$ and $(3/2)^4 > (12/7)^3$. Finally $10^9 < 3^{19}$ gives $10/3 < 9^{5/9}$. This proves (17).

For a uniform loss define

$$S(h,q) = L(rh+q) - \log\left(a^h \beta_*^{q/r} + \beta_*^{-aq/r} \right).$$

Put $u = rL - \log a > 0$, $v = L - \log \beta_*/r > 0$, and $v_0 = L - \log \beta/r > 0$. Differentiation gives $S_h \geq u$ and $S_q \geq v$. With $S_1 = S(1, d-1) > 0$, take

$$\eta = \min\left\{ \frac{u}{r(d+1)}, \frac{v}{r(d+1)}, \frac{S_1}{rd(d-1)}, v_0 \right\} > 0. \tag{18}$$

If $h \geq d$ and $(h,q) \neq (d,0)$, put $t = h - d + q \geq 1$. Then $S(h,q) \geq \min(u,v)t$ and $rh + q \leq r(d+1)t$. If $1 \leq h < d$, concavity of $S(h, d-h) = -\log F(h, d-h)$ gives $S(h, d-h) \geq S_1/(d-1)$. Since

$rh + d - h \le rd$, this pays at least η per vertex at $q = d - h$. Increasing q costs at least $v \ge \eta$ per additional vertex, so $S(h, q) \ge \eta(rh + q)$. Each residual vertex contributes $v_0 \ge \eta$.

The only unpenalized component has $h = p = d$, so its bipartite graph is $K_{d,d}$. All page degrees are used, making it an actual reference component of H. Every other component vertex and the z residual vertices constitute exactly $b(H)$. Inserting their losses in (16) proves (4) and its equality statement. □

We next preserve this loss after deleting dual vertices. The completion below is designed to keep the primal hypergraph simple.

Lemma 3.2. *Let G be the labeled dual of H, with nd/r vertices and n labeled d-edges. For $U \subseteq V(G)$, let $J = G[V(G) \setminus U]$, retaining whole edges. With $L = \log A/(rd)$ and η from* (18)*,*

$$\mathrm{pm}(J) \le \exp\bigl(nL - \eta b(H) + \eta C_0 |U|\bigr), \qquad C_0 = r^2 d[1 + r(d-1)]. \tag{19}$$

Proof. Suppose J is nonempty and has a perfect matching, so all its degrees are positive. Take d disjoint copies. For each original vertex x, add $r - d_J(x)$ parallel labeled edges joining its d copies. The resulting G' is d-uniform and r-regular. Its dual H' is r-uniform, d-regular, and has $r|V(J)|$ vertices. It is simple. Within one copy, two vertices of G' are distinguished by a new edge if either is deficient; otherwise original simplicity distinguishes them. Across copies they are distinguished by a retained edge, since their retained degrees are positive.

Independent matching choices in the copies show $\mathrm{pm}(J)^d \le \mathrm{pm}(G') = T(H')$. Theorem 1.3 therefore gives

$$\mathrm{pm}(J) \le A^{|V(J)|/d^2} \exp(-\eta b(H')/d).$$

Let R_1 be the original dual edges meeting U, let D_1 be the retained dual vertices whose degrees decreased, and let W_1 be the original dual edges meeting $U \cup D_1$. Then $|D_1| \le r(d-1)|U|$ and $|W_1| \le r[1 + r(d-1)]|U|$.

Every old primal vertex outside W_1 retains all its incident edges in each copy. A new reference component containing a formerly nonreference vertex must meet a retained copy of W_1. Without new vertices, a component not meeting W_1 would already have been a reference component. With a new vertex, an adjacent dual vertex lies in D_1 and has a retained old edge, supplying the required retained copy of W_1.

Deletion removes at most $d|R_1|$ old nonreference copies. There are at most $d(|W_1| - |R_1|)$ new reference components meeting retained W_1 copies, each with rd vertices. Hence $b(H') \ge db(H) - rd^2|W_1|$. Substitution and $|V(J)| \le nd/r$ prove (19). If there is no matching, the claim is trivial. If $V(J)$ is empty, its count is one and $C_0|U| \ge n \ge b(H)$, so the bound remains valid. □

Lemma 3.3. *There exists $\varepsilon(r, d) > 0$ such that, if $b(H) = n$ and $M = (r-1)n/r$, every integer $M - \varepsilon n \le k \le M$ satisfies*

$$i_k(H) \le \exp(nL - \eta n/2). \tag{20}$$

Neither M nor n/r needs to be integral.

Proof. The complement of an independent set of size k is a dual edge cover of size $n/r + s$, where $s = M - k \ge 0$. Its total excess coverage is the integer ds. At most ds vertices are covered more than once. Remove the selected edges meeting them, a collection B of size at most rds. What remains is a matching whose uncovered vertex set U has size $d(|B| - s) \le rd^2 s$. The cover is determined by B, U, and that matching. Lemma 3.2 therefore bounds its count by

$$\left(\sum_{j \le rd\varepsilon n} \binom{n}{j}\right)\left(\sum_{j \le rd^2\varepsilon n} \binom{nd/r}{j}\right) \exp(nL - \eta n + \eta C_0 r d^2 \varepsilon n).$$

Set $K = C_0 r d^2$ and choose

$$\varepsilon = \min\left\{ \frac{1}{2r^2 d}, \frac{1}{4K}, \frac{\eta^2}{64 d(\sqrt{r}+d)^2} \right\}.$$

For the binary entropy $h(t) = -t\log t - (1-t)\log(1-t)$, the two binomial tails have total logarithm at most

$$nh(rd\varepsilon) + (nd/r)h(r^2 d\varepsilon) \le 2n\sqrt{d}(\sqrt{r}+d)\sqrt{\varepsilon} \le \eta n/4.$$

Here both arguments are at most $1/2$ and $h(t) \le 2\sqrt{t}$. The completion cost is at most $\eta n/4$. This proves (20). □

Proof of the large-activity part of Theorem 1.1. Assume $b(H) = n$. The inequality $P(\lambda) \ge A\lambda^{(r-1)d}$ bounds its reference power below by $\mathrm{e}^{nL}\lambda^M$. Lemma 3.3 bounds the ratio contributed by terms with $k \ge M - \varepsilon n$ by $\mathrm{e}^{-\eta n/2}/(1-1/\lambda)$. The remaining sets number at most 2^n, and their ratio is at most $(2\mathrm{e}^{-L}\lambda^{-\varepsilon})^n$.

Choose $\lambda_+ > 1$ so that

$$\frac{1}{1-1/\lambda_+} \le \mathrm{e}^{\eta/4}, \qquad 2\mathrm{e}^{-L}\lambda_+^{-\varepsilon} \le \mathrm{e}^{-\eta/8} - \mathrm{e}^{-\eta/4}.$$

For $\lambda \ge \lambda_+$ the first ratio is at most $\mathrm{e}^{-\eta n/4}$. The sum is at most $\mathrm{e}^{-\eta n/8}$, since $x^n + y^n \le (x+y)^n$ for nonnegative x, y. For general H, remove its reference components. The remainder has $b(H)$ vertices, remains simple and regular, and has no reference component. Apply the result to it and multiply back the reference factors, proving (3). □

We finish the activity comparison by treating degree two. The edge-cover duality, regional entropy bound, and doubled regions below are due to Sarantis, Tetali, and Zheng [8]. Ferudun [4] uses a factor of edges and induced odd cycles to prove the unweighted result for every uniformity. We give the weighted argument in full. The pair identity and the fourth spectral moment handle all positive activities simultaneously.

Proof of Theorem 1.2. The empty hypergraph and $\lambda = 0$ give equality directly. Assume $n > 0$ and $\lambda > 0$, and put $x = 1 + 1/\lambda > 1$. Form the dual multigraph F with one vertex for each edge of H and one edge for each vertex of H, joining its two incident hyperedges. It is loopless and r-regular. Every edge multiplicity is at most $r-1$, since multiplicity r would make two hyperedges identical. Write $N = |V(F)|$, so that $n = rN/2$.

Let S be the simple support graph of F. Give a support edge of multiplicity m weight $x^m - 1$, and write $E_x(F)$ for the sum of the products of edge weights over all edge covers of S. The complement of an independent set of H corresponds to an edge cover of F. Summing the nonempty choices in each parallel class gives

$$Z_H(\lambda) = \lambda^n E_x(F). \tag{21}$$

Define

$$T_r(x) = x^{2r} - 4x^r + 2x + 2x^{r-1} - 1.$$

The dual of $H_{r,2}$ is a four-cycle with alternating multiplicities $r-1, 1$. Its weighted cover sum is

$$x^2(x^{r-1}-1)^2 + (x-1)^2(2x^{r-1}-1) = T_r(x).$$

Hence $\lambda^{2r}T_r(x) = Z_{H_{r,2}}(\lambda)$. By (21), it suffices to prove $E_x(F) \le T_r(x)^{N/4}$ and characterize equality.

Partition $V(S)$ into regions B. A boundary state ω_B specifies which edges between B and its complement are selected. Let $w(\omega_B)$ be their weight product, and let $V_B(\omega_B)$ be the vertices

of B covered by them. For $U \subseteq B$, let $Z_B(U)$ be the total weight of internal edge sets covering U, with no restriction on coverage outside U. Put

$$A_B(\omega_B) = w(\omega_B)^{1/2} Z_B(B \setminus V_B(\omega_B)), \qquad \Xi_B = \sum_{\omega_B} A_B(\omega_B)^2.$$

Each boundary edge occurs in two regions. Summing first over the internal edges gives the exact identity $E_x(F) = \sum_\omega \prod_B A_B(\omega|_{\partial B})$. Generalized Hölder, or its entropy proof, yields

$$E_x(F) \le \prod_B \Xi_B^{1/2}. \tag{22}$$

For completeness, normalize the summands in the identity to a probability law μ on boundary states. Shearer's entropy inequality gives $H(\mu) \le \frac{1}{2}\sum_B H(\mu_B)$, because each boundary coordinate occurs twice. For each marginal, the variational inequality gives

$$\mathbb{E}_{\mu_B} \log A_B + \tfrac{1}{2} H(\mu_B) \le \tfrac{1}{2} \log \sum_{\omega_B} A_B(\omega_B)^2.$$

Summing proves (22). Zero terms are omitted from the support. This argument uses arbitrary nonnegative edge weights.

For a two-vertex region $B = \{u, v\}$ whose internal edge has multiplicity m, set $a = x^m - 1$ and $h = x^{r-m} - 1$. Grouping states according to which endpoints are covered by boundary edges gives

$$\Xi_B = a^2 + 2ha^2 + h^2(1+a)^2 = x^{2r} - 4x^r + 2(x^m + x^{r-m}) - 1.$$

Consequently,

$$T_r(x) - \Xi_B = 2(x^{m-1} - 1)(x^{r-m} - x) \ge 0. \tag{23}$$

If S has a perfect matching, use its edges as the regions. Equation (22) already proves the desired bound in this case.

For general S, the multiplicities divided by r form a fractional perfect matching. An extreme point of its polytope has a support consisting of single edges and odd cycles. Indeed, the positive columns of the vertex-edge incidence matrix are linearly independent. A degree-one vertex in the support forces its incident weight to be one and hence forms an isolated edge. Every remaining component has minimum degree at least two and at most as many edges as vertices, so it is a cycle; an even cycle would give a linear dependence. Thus an edge-and-odd-cycle factor exists.

Choose such a factor with the fewest vertices in odd cycles. Its cycles are induced in S. A chord of an odd cycle would form a shorter odd cycle with one of the two paths between its endpoints, while the internal vertices of the other path could be paired along the original cycle. This would reduce the number of vertices in odd cycles.

Use the edges and cycles of this factor as regions. It remains to bound the regions that are induced odd cycles. Suppose first that a cycle region B has a boundary. For $v \in B$, let b_v be its total boundary multiplicity; then $b_v \le r - 2$. Take two copies of $F[B]$ and join corresponding vertices by b_v parallel edges, omitting such edges when $b_v = 0$. Call the resulting multigraph F_B^*. It is loopless, r-regular, and has maximum multiplicity at most $r - 1$. Grouping boundary selections by the set D of covered vertices gives

$$\Xi_B = \sum_{D \subseteq B} \prod_{v \in D} (x^{b_v} - 1) Z_B(B \setminus D)^2 = E_x(F_B^*).$$

Since some $b_v > 0$, the support of F_B^* has a perfect matching. Use one corresponding edge vv' and match the remaining even paths in the two copies. The already proved perfect-matching case gives

$$\Xi_B \le T_r(x)^{|B|/2}. \tag{24}$$

If B has no boundary, inducedness makes it a whole component. The regularity equations around its odd cycle force r to be even and every edge multiplicity to equal $r/2$. Set $w = x^{r/2} - 1 > 0$ and let $\ell = |B|$ be odd. The transfer matrix

$$K = \begin{pmatrix} 0 & w \\ 1 & w \end{pmatrix}$$

encodes cyclic selections of edges with no two successive omissions, so $E_x(F[B]) = \operatorname{tr}(K^\ell)$. Its eigenvalues satisfy $t > 0 > u$ and $|u| < t$. Therefore

$$E_x(F[B]) = t^\ell + u^\ell < t^\ell < (t^4 + u^4)^{\ell/4}.$$

The fourth moment is

$$t^4 + u^4 = w^4 + 4w^3 + 2w^2 = x^{2r} - 4x^r + 4x^{r/2} - 1 \le T_r(x),$$

since $x + x^{r-1} \ge 2x^{r/2}$. Thus $\Xi_B = E_x(F[B])^2 < T_r(x)^{|B|/2}$. Combining this estimate, (23), and (24) in (22) proves $E_x(F) \le T_r(x)^{N/4}$.

We prove the equality condition first when S has a perfect matching, using its edges as regions. All values of every A_B are positive. There is a direct proof of (22) by successive Cauchy–Schwarz inequalities on boundary variables, replacing each of the two functions containing a variable by its partial square-sum norm. Any boundary variable can be treated first. Equality in the final bound therefore requires equality in this first Cauchy–Schwarz inequality for every fixed state of the other variables.

Let a boundary edge $e = ac$ join the regions $B = \{a, b\}$ and $C = \{c, d\}$. Write $p_B = x^{m_B} - 1 > 0$ for the internal edge weight of B and $w_e > 0$ for the weight of e. For each fixed state of the other boundary edges,

$$\frac{A_B(e = 1)}{A_B(e = 0)} = \sqrt{w_e}(1 + 1/p_B)^{I_B},$$

where $I_B = 1$ precisely when some boundary edge at b is selected and no boundary edge at a other than e is selected. Define I_C by exchanging (a, b) for (c, d). Cauchy–Schwarz equality forces the two ratios to agree. Since both bases exceed one, the Boolean functions I_B and I_C must be identical.

Each vertex has positive boundary multiplicity, because $m_B, m_C \le r - 1$. In a Boolean function requiring at least one positive variable to be selected and all negative variables to be absent, every listed variable is essential, and its influence has the indicated sign. The positive sets for I_B, I_C must therefore coincide, as must the negative sets. The latter consist of support edges at a, c, respectively, other than e. They are disjoint, so both are empty. The positive sets consist of edges at b, d; their intersection can only be bd, so both are the singleton $\{bd\}$. The four vertices consequently form a whole support component with edges ab, ac, cd, bd.

Regularity forces the multiplicities around this four-cycle to alternate between m and $r - m$. Equation (23) is an equality only when $m = 1$ or $m = r - 1$. This proves that, in the perfect-matching case, equality requires a disjoint union of the reference four-cycles, and the displayed reference calculation proves the converse.

Finally return to the edge-and-odd-cycle factor. A closed odd cycle gives a strict inequality by the spectral argument. For a cycle with a boundary, F_B^* is connected, has a perfect matching, and has at least six vertices. It cannot be a union of reference four-cycles, so (24) is strict by the equality condition just proved. Thus equality excludes all cycle regions and reduces to the perfect-matching case. Translating the reference four-cycles back through the duality gives exactly the disjoint unions of $H_{r,2}$. □

4 Coefficient intervals and stability

Partition-function comparison on a real interval does not imply coefficientwise comparison. At small activity we use the analytic control of Lemma 2.6, while at large activity a positive-coefficient recurrence suffices. The Fourier estimate below concerns only the fixed reference polynomial and remains uniform as the activity tends to zero.

Lemma 4.1. *Let P be a fixed nonnegative-coefficient polynomial with $P(0) = 1$ and $P'(0) = B > 0$. Fix $\lambda_0 > 0$. For $0 < \lambda \le \lambda_0$, let $\mathbb{P}(Y = j) = [z^j]P(z)\lambda^j/P(\lambda)$. Suppose q is a positive integer and $k = q\mathbb{E}Y$ is integral. Define*

$$\Phi(\theta) = \left(\frac{P(\lambda e^{i\theta})}{P(\lambda)}\right)^q, \qquad p_k = \frac{1}{2\pi}\int_{-\pi}^{\pi} \Phi(\theta) e^{-ik\theta}\, d\theta,$$

and let J_0, J_1 be the integrals of $|\Phi|$ and $|\theta\Phi|$, with the same factor $1/(2\pi)$. Uniformly for $0 < \lambda \le \lambda_0$, as $k \to \infty$,

$$p_k \sim (2\pi q \operatorname{Var} Y)^{-1/2}, \qquad J_0/p_k \longrightarrow 1, \qquad J_1/p_k = O(k^{-1/2}).$$

Also $q \operatorname{Var} Y \asymp k \asymp q\lambda$, and $|\Phi(\theta)| \le \exp[-ck(1 - \cos\theta)]$ for $c > 0$ depending only on P, λ_0.

Proof. The probabilities at zero and one imply

$$|\Phi(\theta)| \le \exp\left[-\frac{qB\lambda}{P(\lambda)^2}(1 - \cos\theta)\right]. \tag{25}$$

Indeed the squared single-block characteristic modulus is at most $1 - 2B\lambda(1 - \cos\theta)/P(\lambda)^2$. Finite support and continuity give $\mathbb{E}Y \asymp \lambda$, $\operatorname{Var} Y \asymp \lambda$, and $\mathbb{E}|Y - \mathbb{E}Y|^3 = O(\lambda)$ throughout the fixed interval. Near zero,

$$\log \Phi(\theta) - ik\theta = -q \operatorname{Var} Y\, \theta^2/2 + O(k|\theta|^3).$$

On a sufficiently small fixed angular interval, half the Gaussian exponent absorbs the real part of the remainder. The central integral differs from its Gaussian approximation by at most $Ck \int |\theta|^3 e^{-ck\theta^2}\, d\theta = O(k^{-1})$. The same applies to the modulus integral. Outside that interval, (25) is exponentially small. It also gives $J_1 = O(k^{-1})$. These estimates prove every assertion uniformly, including $\lambda \to 0$. □

Proof of the low-density part of Theorem 1.4. Fix r, d, write $B = rd$, $\ell = 2r$, and use Lemma 2.6. Choose a sufficiently small $\lambda_0 > 0$ depending only on r, d. With $S = Q(\lambda)$ and $g = f(\lambda)$, equation (14) gives

$$\frac{3}{4}S \le g \le \frac{5}{4}S, \quad \sum_j |[z^j]f|\lambda^j \le 2g, \quad \sum_j j|[z^j]f|\lambda^j \le 2(\ell + 1)g. \tag{26}$$

In addition $g \le Cn\lambda^{r+1}$. Reduce λ_0 so that $P(\lambda_0) \le 4/3$ and $\lambda_0 \le 1/(4B)$. Put $\theta_0 = 1/[4(\ell+1)]$ and impose the further smallness requirements on $C\lambda_0^r$ used below. Each can be met by a positive choice.

Let $q = n/B$ and $\alpha_- = \lambda_0 P'(\lambda_0)/(BP(\lambda_0))$. The mean $\mathbb{E}_\lambda Y$ strictly increases, with derivative $\operatorname{Var}(Y)/\lambda > 0$. Thus each integer $0 < k \le \alpha_- n$ determines a unique $\lambda \in (0, \lambda_0]$ such that $q\mathbb{E}_\lambda Y = k$. Coefficient extraction at that activity gives

$$\frac{i_k(H)\lambda^k}{P(\lambda)^q} = \frac{1}{2\pi}\int_{-\pi}^{\pi} \Phi(\theta) e^{-ik\theta - f(\lambda e^{i\theta})}\, d\theta. \tag{27}$$

If $D = 0$ the conclusion is equality. Otherwise $S, g > 0$, and (26) gives on the whole circle

$$|f(\lambda \mathrm{e}^{i\theta})| \le 2g, \qquad |f(\lambda \mathrm{e}^{i\theta}) - g| \le 2(\ell+1)g|\theta|.$$

In particular $\Re f \ge g/2$ on $|\theta| \le \theta_0$.

If $0 < g \le 1$, the difference between $1-\mathrm{e}^{-f(\lambda \mathrm{e}^{i\theta})}$ and $1-\mathrm{e}^{-g}$ has modulus at most $2(\ell+1)\mathrm{e}^2 g|\theta|$. Subtracting (27) from the reference integral yields a difference at least $(1-\mathrm{e}^{-g})p_k - 2(\ell+1)\mathrm{e}^2 g J_1$. Since $1-\mathrm{e}^{-g} \ge g/2$, Lemma 4.1 makes this at least $gp_k/4$ whenever $k \ge K_0(r,d)$.

If $g \ge 1$, the central part has modulus at most $\mathrm{e}^{-g/2}J_0$. On $|\theta| \ge \theta_0$, equation (25) and $P(\lambda) \le 4/3$ give

$$|\Phi(\theta)\mathrm{e}^{-f(\lambda \mathrm{e}^{i\theta})}| \le \exp[-(9/16)n\lambda(1-\cos\theta) + 2g] \le \exp[-n\lambda(1-\cos\theta_0)/4].$$

The second inequality follows from $g \le Cn\lambda^{r+1}$ by reducing λ_0. Since P is coefficientwise at most $(1+z)^B$, one has $k \le en\lambda$; since $P'(\lambda) \ge B$, one also has $k \ge 3n\lambda/4$. The outer integral divided by p_k is consequently $O(\sqrt{k}\mathrm{e}^{-ck})$, with $c = (1-\cos\theta_0)/(4\mathrm{e})$. Reduce λ_0 further so that $g/4 \le ck/2$. For sufficiently large K_0, Lemma 4.1 then yields

$$\frac{i_k(H)}{i_k(H^n_{r,d})} \le \mathrm{e}^{-g/4} \le \mathrm{e}^{-3Q(\lambda)/16} \qquad (K_0 \le k \le \alpha_- n). \tag{28}$$

Indeed, after division by $\mathrm{e}^{-g/4}$ the central contribution is at most $(11/10)\mathrm{e}^{-1/4} < 9/10$, and the outer one is at most $1/10$. The first case already gives $1 - g/4 \le \mathrm{e}^{-g/4}$. This retains the deficit even when it is arbitrarily small.

We finish the bounded indices directly from the layered expansion. For fixed $k < K_0$, put $W_k = \sum_{j\in\mathcal{J},\, j\le k} D_j \binom{n}{k-j}$. This is the coefficient of $(1+z)^n Q(z)$. Each polynomial error of Lemma 2.6 is controlled by a deficit whose main degree is strictly smaller, so it contributes $O_{r,d,k}(W_k/n)$. The tail starts at $\ell+1$ and has bounded coefficients times D; when $k > \ell$ the same conclusion follows from $W_k \ge D\binom{n}{k-\ell}$ for sufficiently large n, and otherwise it contributes zero.

The exponential conversion has smaller error. Put $h(z) = \log Z_H(z) - n\log(1+z)$. Lemma 2.5 gives coefficient bounds $h = O_{r,d}(nz^r)$ on a fixed disk. In the exact identity

$$Z_{H^n_{r,d}} - Z_H = (1+z)^n \mathrm{e}^{h(z)}(\mathrm{e}^{f(z)} - 1),$$

every term beyond the linear term f has another factor h, beginning at degree r, or another factor f, beginning at degree $r+1$. Since $D = O_{r,d}(n)$ and k is fixed, these contributions are $O_{r,d,k}(n^{1-r}W_k)$. If $W_k = 0$, all deficits with main degree at most k vanish, so f has no coefficient through degree k and the two independent-set coefficients agree. Otherwise their difference is $W_k(1 + O_{r,d,k}(1/n)) > 0$ for sufficiently large n. A common $N(r,d)$ handles the finitely many indices below K_0. For $k \ge 2r$, nonreference graphs have $W_k > 0$, proving the asserted equality classification.

Finally if $k \ge \tau n$, then $\lambda \ge \tau/\mathrm{e}$. Since $\lambda \le 1$, one has $Q(\lambda) \ge D\lambda^{2r}$, while $D \ge \mathrm{dist}(H, H^n_{r,d})/C$. Equation (28) proves (5). Thus a logarithmic coefficient deficit of $o(n)$ on a fixed positive density band forces edge-edit distance $o(n)$. □

For the other end, reverse the reference polynomial by setting $Q_+(t) = t^{(r-1)d}P(1/t)$ and $F(t) = Q_+(t)/A$. Then $F(0) = 1$ and $u = F'(0) > 0$. A weighted double count controls adjacent coefficients of powers of this fixed polynomial.

Lemma 4.2. *If $a_j = [t^j]Q_+(t)^g$ and $g \ge 1$, then for $1 \le j \le g$,*

$$a_j \ge \frac{u(g-j+1)}{j} a_{j-1}. \tag{29}$$

Proof. A weighted choice of one monomial from each factor, with total degree $j-1$, has at least $g-j+1$ constant coordinates. Change one from A to the linear term Au, multiplying its weight by u. Each resulting degree-j choice is counted at most j times, once per linear coordinate. Summation proves (29). □

Proof of the high-density part of Theorem 1.4. Retain $B = rd$, $A = 1 + (r-1)^d$ and $\eta, \varepsilon, \lambda_+$ from Section 3. Put $c = \eta/8$, $D_+ = F(1) - 1 > 0$. Nonnegative coefficients give $F(t) \le 1 + D_+ t$ and $\log F(t) \le D_+ t$ on $0 \le t \le 1$. Choose

$$\rho = \min\{1/\lambda_+, 1/2, cB/(2D_+)\}, \qquad t_* = \min\{1/2, \eta B/(8D_+)\},$$

and set

$$\alpha_+ = \min\left\{\frac{1}{4B}, \frac{u\rho}{4B}, \frac{\varepsilon}{2}, \frac{\eta}{16\log(1/t_*)}\right\} > 0. \tag{30}$$

Remove all reference components and write $H = H_{r,d}^{\sqcup g} \sqcup G$, $b = |V(G)| = b(H)$, $v = b/B$, and $m = n/B = g + v$. Divisibility makes v integral. The case $b = 0$ is equality, so suppose $b > 0$. The polynomial $R_G(t) = t^{(r-1)dv} Z_G(1/t)$ has nonnegative coefficients and no negative powers, since every independent set of G has size at most $(r-1)b/r$. For $s = (r-1)n/r - k$,

$$\frac{i_{(r-1)n/r-s}(H)}{i_{(r-1)n/r-s}(H_{r,d}^n)} = \frac{[t^s]Q_+(t)^g R_G(t)}{[t^s]Q_+(t)^m}. \tag{31}$$

The case $s = 0$ follows from Theorem 1.3, so assume $1 \le s \le \alpha_+ n$.

If $b < n/2$, then $g > n/(2B)$ and $s \le g/2$. Put $a_j = [t^j]Q_+^g$ and $x = s/[u(g-s+1)] \le 4B\alpha_+/u \le \rho$. Iterating (29) gives $a_{s-j}/a_s \le x^j$ for $0 \le j \le s$. The denominator in (31) is at least $A^v a_s$, so the ratio is at most

$$R_G(x)/A^v \le \mathrm{e}^{-cb} F(x)^v \le \exp[-cb + (b/B)D_+ x] \le \mathrm{e}^{-\eta b/16}.$$

Here Theorem 1.1 is applied to G at activity $1/x \ge \lambda_+$. Its use for coefficients is justified by the recurrence, not by coefficientwise comparison of functions.

If $b \ge n/2$, then $s \le 2\alpha_+ b \le \varepsilon b$. Lemma 3.3 bounds each $[t^j]R_G$, $j \le s$, by $A^v \mathrm{e}^{-\eta b/2}$. Hence the numerator in (31) is at most $A^v \mathrm{e}^{-\eta b/2} \sum_{j=0}^s a_j$, allowing $g = 0$.

The denominator is at least A^m. Indeed, if $M = (r-1)n/r$, deletion of s vertices from maximum independent sets gives

$$i_{M-s}(H_{r,d}^n) \ge i_M(H_{r,d}^n) \frac{\binom{M}{s}}{\binom{n-M+s}{s}} \ge A^m.$$

The last inequality uses $s \le 2M - n$, ensured by (30). Also $\sum_{j=0}^s a_j \le t_*^{-s} Q_+(t_*)^g$. Since $g \le b/B$ and $s \le 2\alpha_+ b$, the ratio's logarithm is at most

$$-\eta b/2 + (b/B)D_+ t_* + 2\alpha_+ b \log(1/t_*) \le -\eta b/4.$$

Both cases imply (6). The inequality is strict for $b > 0$ and is equality for reference unions. A logarithmic deficit of $o(n)$ in this interval consequently forces $b(H) = o(n)$. □

For degrees at least three, the two intervals leave a middle range untreated. At large activity, the proof budgets covers only within a small linear excess of the minimum size. At small activity, later terms must remain subordinate to the local structural deficits. These are limitations of the arguments; no counterexample to either inequality in (2) is obtained here.

Data availability

The accompanying `supplement` contains exact checks of the degree-two formulas, with instructions for reproducing them. These checks support the analytic proofs and are not used in place of them. GPT-6.1 (OpenAI) assisted with implementing the verification programs.

Declaration of AI use

GPT-6.1 assisted with exploring proof strategies, checking calculations, and preparing and revising the exposition. The author is responsible for the mathematical claims and the final version of the manuscript.

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